

Homework (Carolina Reaper)

- Q1.** A painter creates 5 canvases with sizes (in cm) 20, 35, 40, 45, 50. What is the first quartile (Q_1) of the canvas sizes?
(A) 20
(B) 30
(C) 35
(D) 40
(E) 45
- Q2.** A music teacher records the number of rhythm patterns her students can play: 12, 15, 18, 20, 22, 25, 30. What is the interquartile range (IQR)?
(A) 5
(B) 6
(C) 8
(D) 10
(E) 12
- Q3.** What is the Q_3 for the dataset 10, 15, 20, 25, 30, 35, 40, 45?
(A) 25
(B) 27.5
(C) 30
(D) 32.5
(E) 35
- Q4.** Identify the outlier in the dataset 23, 25, 26, 27, 28, 120.
(A) 23
(B) 25
(C) 26
(D) 27
(E) 120
- Q5.** What is the lower bound for outliers in a box plot for the dataset 10, 12, 15, 18, 22, 25 if $Q_1 = 12$, $Q_3 = 22$?
(A) 5
(B) 7
(C) 9
(D) 10
(E) 11
- Q6.** Determine the number of data points needed to construct a box-and-whisker plot for five rhythm patterns with 4, 6, 8, 10, 12 beats.
(A) 5
(B) 6
(C) 10
(D) 12
(E) 15
- Q7.** What does $IQR = 12$ indicate about the variation in the dataset?
(A) Small variation
(B) Large variation
(C) No variation
(D) Cannot be determined
(E) Equal variation
- Q8.** If the median (Q_2) of a box plot is x and $Q_1 = 2x$, what is Q_3 ?
(A) $2x$
(B) $3x$
(C) $4x$
(D) $5x$
(E) $6x$

Open Ended Questions (FRQ)

- Q9.** Create a box plot for the dataset 5, 8, 12, 15, 18, 20, 25. Label Q_1 , Q_2 , Q_3 , and identify any outliers.
- Q10.** Explain how to determine the interquartile range (IQR) from a box plot, using the dataset 10, 12, 15, 18, 20, 22, 25 as an example.

Chef's Tips

- Tip 1: Ensure students use correct units and labels.
- Tip 2: Check for accurate calculation of quartiles and IQR.
- Tip 3: Verify proper identification of outliers.